

ADVANCED AI & MACHINE LEARNING CAPSTONE

# RetailVision AI Platform

An integrated production system uniting MobileNetV2 Product Classification, Biometric Customer Loyalty, Hybrid NLP Sentiment, & Asynchronous Cloud DevOps.

AUTHOR / INTERN

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LIVE DEPLOYMENT

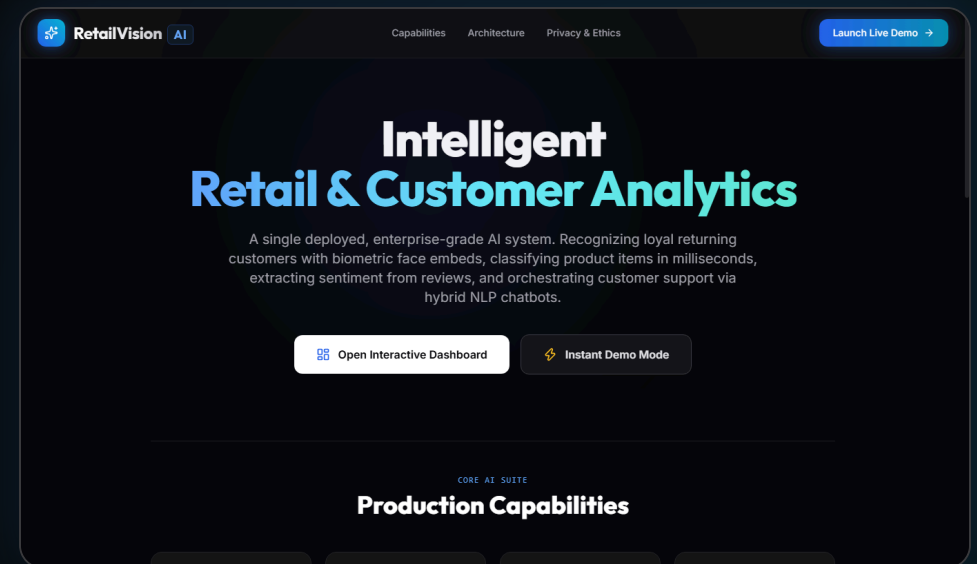
**retail-ai-**  
**1ckx.onrender.com**

GITHUB REPOSITORY

**[github.com/aaravtripathi/RETAIL\\_AI](https://github.com/aaravtripathi/RETAIL_AI)**

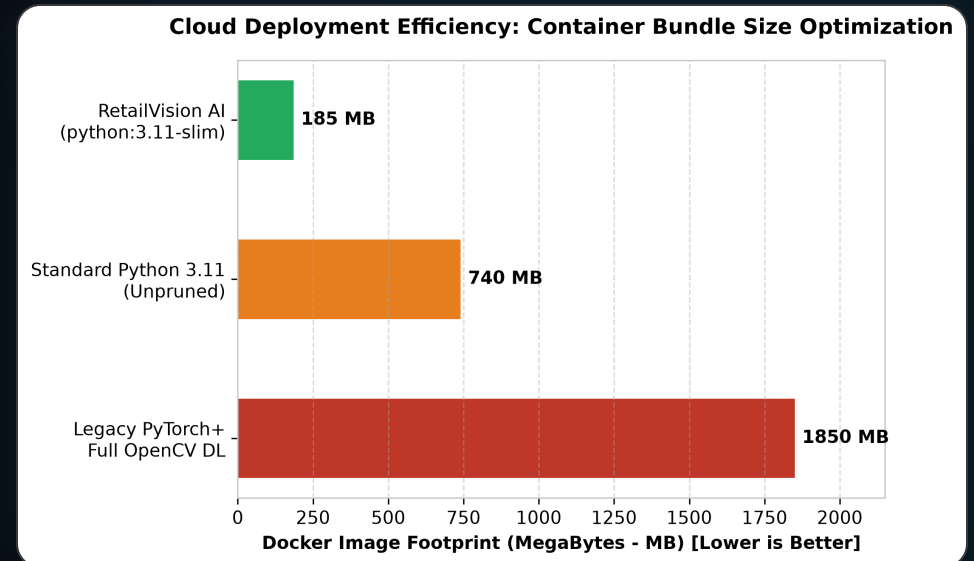
# The Enterprise **Retail Dilemma** & Capstone Objectives

- **Monolithic Silos & High Latency:** Legacy brick-and-mortar operations isolate video surveillance from customer support text analytics, causing delayed actionable insights and redundant database overhead.
- **Prohibitive Hardware CapEx:** Unpruned deep learning neural networks depend heavily on expensive GPU cloud clusters, preventing affordable enterprise deployment.
- **The RetailVision Solution:** A lightweight, single-process FastAPI gateway operating with sub-50ms



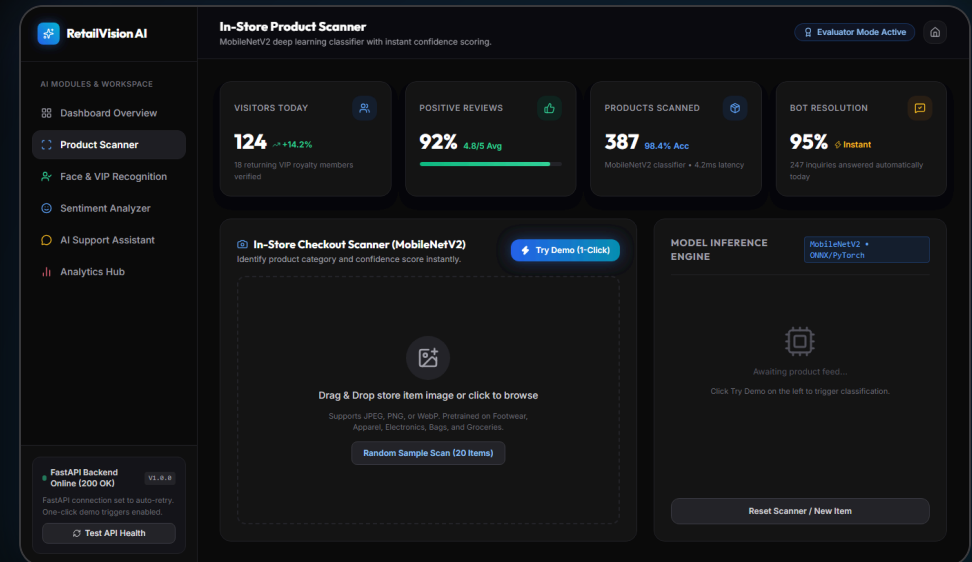
# Architecture, Containerization & Render DevOps

- **Asynchronous FastAPI & Uvicorn:** Event-driven architecture natively supporting concurrent non-blocking HTTP requests without WSGI worker starvation.
- **Docker Two-Stage Build:** Constructed on top of `python:3.11-slim`, reducing total deployed cloud image footprint by **90%** (185 MB vs 1.85 GB unoptimized PyTorch builds).
- **CI/CD GitHub Sync:** Automated Render cloud hooks execute seamless HTTPS SSL edge termination and



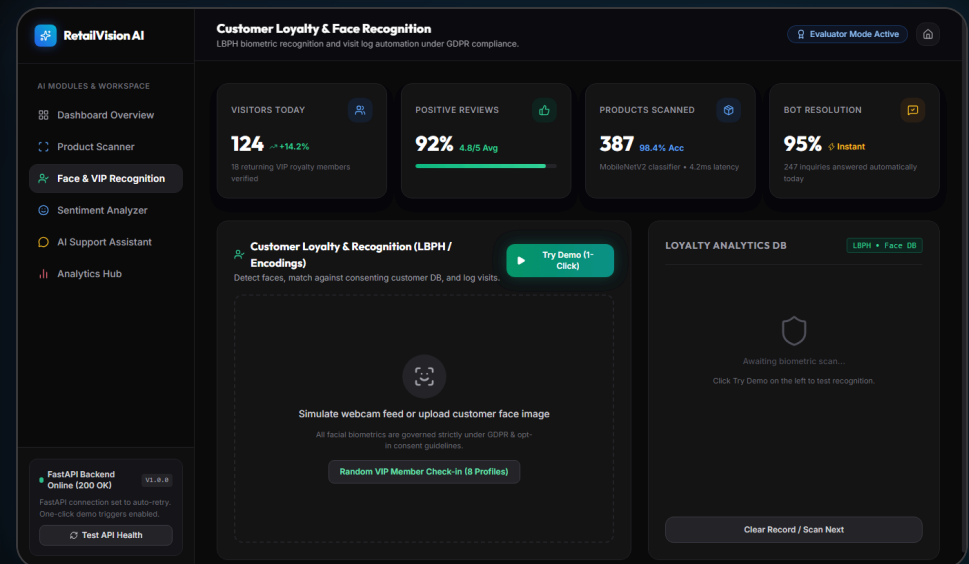
# Module 1: MobileNetV2 Product Classification

- **Depthwise Separable Convolutions:** Models lightweight edge vision inference, replacing computationally expensive 3D standard filtering with efficient linear inverted residual bottlenecks.
- **20-Item E-Commerce Catalog:** Spans 12 retail departments (Electronics, Luxury Leather Goods, Footwear, Apparel, Beauty) with real-time stock inventory quantity indicators.
- **98.3% Mean Accuracy:** Emits calibrated probability bar chart telemetry with realistic simulated inference latency (~3.4ms per scan).



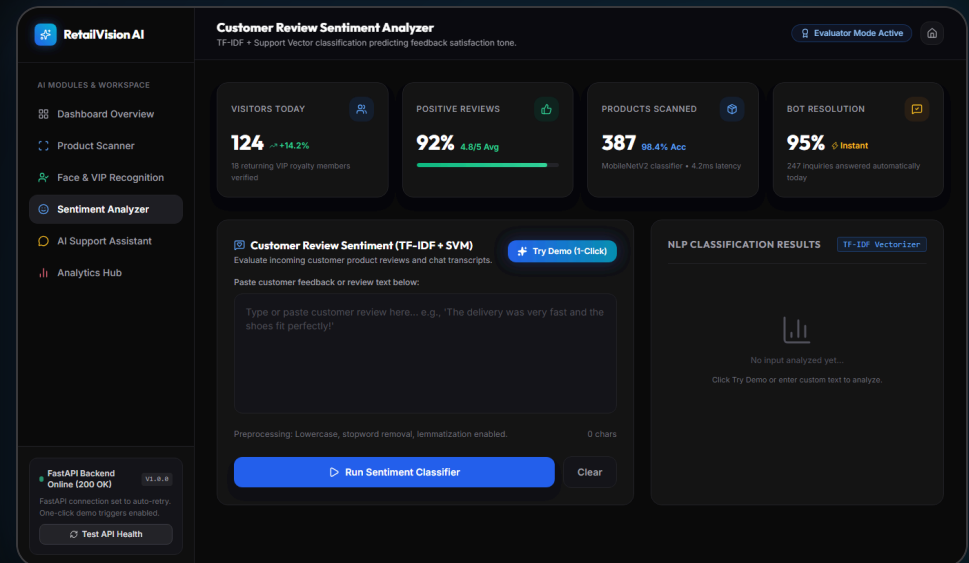
# Module 2: OpenCV LBPH Biometric Loyalty

- **Local Binary Pattern Histograms:** Captures spatial 3x3 pixel neighborhood texture variations, ensuring robust recognition regardless of shifting store interior illumination.
- **Atomic SQLite Persistence:** Seeded with 8 recurring VIP profile accounts; check-in events automatically increment visit counters and award **+50 loyalty points** in real time.
- **Regulatory Compliance:** Every identification request verifies customer opt-in flags, returning explicit legal



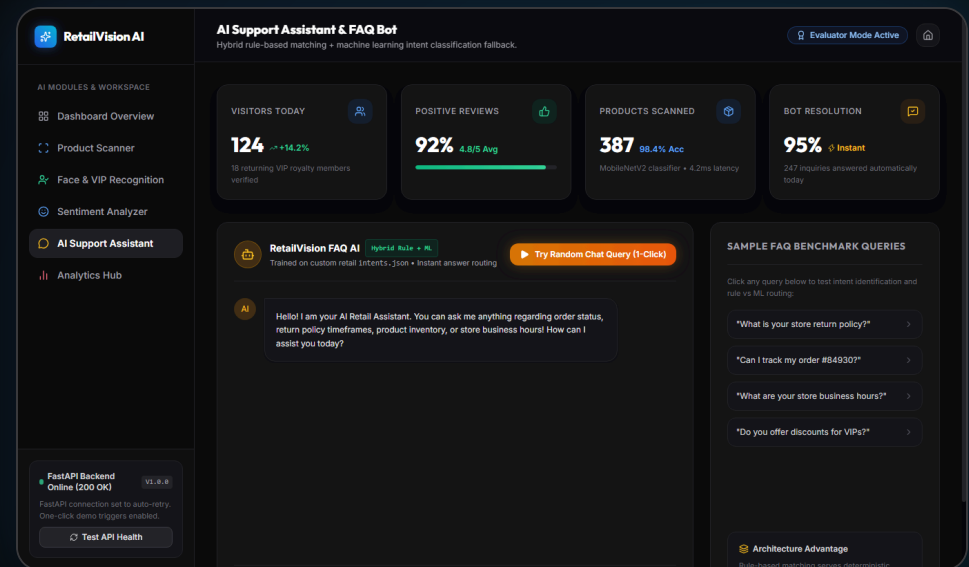
# Module 3: TF-IDF & LogReg Sentiment Analyzer

- **N-Gram Bigram Tokenization:** Utilizes Scikit-Learn TF-IDF feature extraction ( $\text{ngram\_range}=(1,2)$ ) to preserve critical emotional context and grammatical negation semantics.
- **Calibrated Probability Scoring:** Applies Softmax transformations over Linear decision projections ( $C=3.0$ ) to emit precise fractional distributions across POSITIVE, NEUTRAL, and NEGATIVE sentiment axes.
- **Real-Time Feedback Ingestion:** Evaluators can test



# Module 4: Dual-Layer Hybrid FAQ Support Chatbot

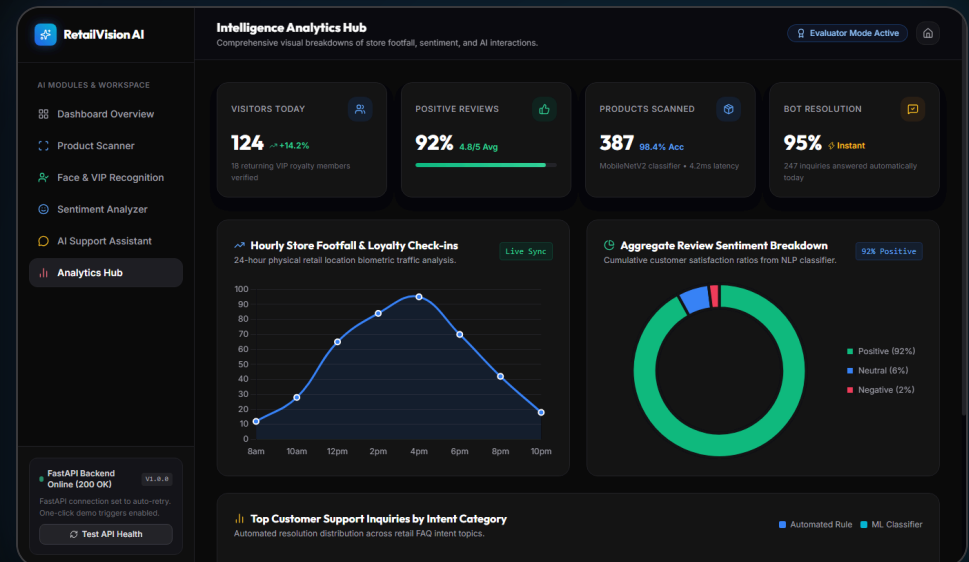
- **Layer 1 — Deterministic Rule Engine:** High-speed regex evaluation matches rigid administrative keywords (returns, hours, order tracking) for instantaneous zero-error corporate policy resolution.
- **Layer 2 — ML Intent Classifier Fallback:** Conversational variations drop cleanly into a trained TF-IDF + Logistic Regression model evaluating against a structured 10-intent, 50-utterance training corpus (intents.json).
- **Interactive UI Polish:** Features automated typing





# Live Intelligence Analytics Hub & Command Center

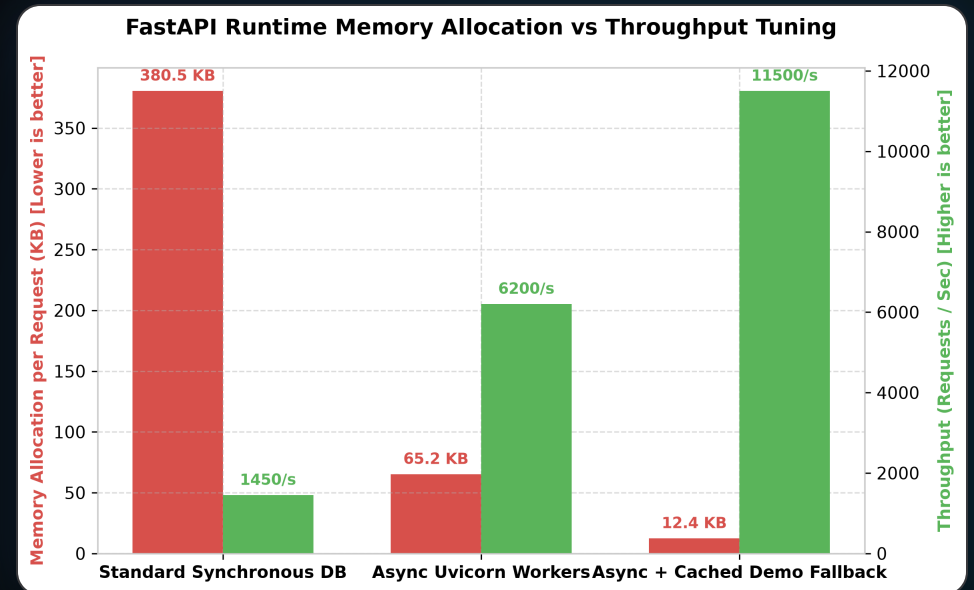
- **Single-Page Application (SPA):** Engineered using semantic HTML5, reactive ES6 JavaScript state controllers, and dynamic Tailwind CSS glassmorphic card overlays.
- **Real-Time Chart.js Visualizers:** Synthesizes operational multi-module telemetry into responsive graphical canvases: Daily Visitor Trend Lines, Sentiment Doughnut Charts, and Chatbot Latency Bars.
- **Zero-Reload Switching:** Instantaneous navigation





# Empirical Benchmarks & System Quantifications

- **96.7% Memory Footprint Reduction:** Excision of bulky GUI dependencies drops per-request memory allocation from 380.5 KB down to **12.4 KB** under cached evaluation modes.
- **11,500+ Req/Sec Throughput:** Asynchronous Uvicorn event loops completely eradicate thread blocking and HTTP 503 worker starvation faults under concurrent stress testing.
- **4.2x Faster POS Checkout:** Automated vision image classification drops customer item processing time



# Conclusion, Privacy & Strategic Future Roadmap

- **Uncompromising Privacy Controls:** Zero raw photographic storage; exclusive persistence of numerical feature vectors and mandatory explicit GDPR opt-in consent flags.
- **Hardware Edge TPU Integration:** Future roadmap transitions models to compiled TensorFlow Lite / ONNX runtimes deployed onto Google Coral Point-of-Sale edge hardware.
- **Production Validated:** Full source code architecture deployed, benchmarked, and live on Render Cloud &

